

Directional Collapse in Fine-Tuning: Preserving the Way Back

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Abstract

Training on one direction of a paired task can improve forward performance while damaging an existing ability to perform the backward task. We study this tradeoff across code transformation, format conversion, translation, semantic parsing, factual relations, execution prediction, and algebra. Our preservation baseline replaces a small fraction of forward training pairs with their reversals, using ordinary cross-entropy from the outset. In formatting, 1% reversal yields 90.2% backward success versus 0.4% after forward-only training, while forward success is 99.6% versus 99.7%. Code, translation, and semantic parsing also show substantial backward benefits, with varying forward costs and required doses. Replacement fixes training-example counts and optimizer steps and requires no new paired corpus. Comparisons against contrastive and other auxiliary objectives test what those objectives add beyond directional supervision; mixed-direction losses sometimes add benefits. Likelihood and layer-intervention diagnostics suggest output-selection interference as a supporting explanation. Adverse algebra doses and a repeated unit-conversion null delimit the result. Small-dose reversed-pair SFT is a necessary baseline for attributing preservation to specialized objectives.

1 Introduction

A translation pair can train either language direction. A question–SQL pair can train semantic parsing or question generation. Although the same pair supplies both tasks, training on one direction need not preserve the other. Prior work documents directional degradation in code transformation (Nikiema et al., 2025) and translation (Zhu et al., 2024). Evaluating only the trained direction can therefore miss a substantial loss of an existing capability.

The practical intervention is simple: replace some forward examples with their reversed pairs

and train with ordinary cross-entropy. In our formatting comparison, forward-only SFT improves forward success from 60.2% to 99.7%, but backward success falls from 51.1% to 0.4%. With 1% reversal, backward success is 90.2% and forward success is 99.6%. The mixture is trained from the base checkpoint; it prevents much of the backward loss rather than repairing an already collapsed model with additional data.

This baseline changes what a specialized preservation objective must demonstrate. An auxiliary objective may change directional information as well as optimization. Encoding both sides of a pair, or adding backward prediction targets, can help independently of a particular loss. We first measure preservation with reversed-pair SFT, then compare auxiliary objectives against that directional supervision and appropriate exposure controls. Matching examples and optimizer steps does not guarantee equal tokens, runtime, or FLOPs.

The paper makes three connected contributions. First, it characterizes forward/backward tradeoffs across task families, models, and training runs. Second, it measures how those tradeoffs change with the proportion of reversed pairs. Third, it tests whether specialized objectives add benefits beyond ordinary directional supervision. The required dose varies by task: low doses can preserve backward performance with little forward change, but they can also impose costs or fail. Unit conversion provides a repeated case where forward SFT improves both directions. Supporting diagnostics examine output selection; they do not establish memorization, erased knowledge, or a universal internal mechanism.

2 Paired tasks and directional degradation

A paired instance contains two sides (x, y) , together with any required context such as a func-

tion or database schema. The forward task generates y from x ; the backward task generates a valid x from y . Reversing a paired task need not mean uniquely reconstructing a string: translations and questions admit paraphrases, and a function can have several inputs producing the same output. Correctness follows the requested task, rather than reference-string matching whenever a semantic check is available.

We call any reduction in backward success after forward-only training *directional degradation*. For descriptive counting and highlighting, *severe directional collapse* means backward success at most half the base-model rate. This threshold describes the size of a loss; it is not a significance test. All comparisons use the base model evaluated on the same held-out examples as the fine-tuned models.

3 Tasks and experimental setup

What the model must produce. Table 1 defines the seven task families and their correctness checks. Translation contributes four training orientations and algebra two, giving 11 orientations in the main comparison; opposite orientations are not independent datasets. Formatting converts JSON to YAML/XML and Markdown tables to CSV, with the backward tasks converting back. Code pairs readable Python with five obfuscation styles. Execution prediction supplies the function in both directions and accepts any input producing the required backward output.

What success means. Each answer receives a binary, task-specific correctness score; strict success is the percentage satisfying that predicate. Format outputs must parse with the target parser, match the underlying data, and avoid copying the input. Copying matters particularly for JSON-to-YAML: JSON also parses as YAML. The copy guard rejects this trivial behavior, although the original scorer does not require syntax exclusive to YAML. Code correctness primarily measures agreement on execution tests, not recovery of original identifier names. Algebra checks symbolic equivalence; factoring also requires a product form, while the expansion scorer has no separate expanded-form requirement.

Translation requires the target language and COMET-22 above a threshold fixed from each base model’s score distribution before tuned evaluation. Thresholds differ by model and direction,

so success percentages compare training methods within a model, not absolute translation quality across models. The original forward translation check lacks the backward check’s explicit copy rejection. SQL generation uses database execution; backward question generation is scored by parsing the question with a frozen Granite model and checking the resulting query. This model-based check can reject valid paraphrases. The other proposed model-based check, WebNLG triples-to-text, has an unreliable reference extractor and contributes no tuned result. Exact prompts, parsing rules, thresholds, source versions and split sizes are consolidated in Appendix A.

Training data and intervention. Table 1 prominently gives retained training sizes. In the reference formatting run, 1% means 65 reversed pairs among 6500; in code it means 62 among 6499. The actual integer counts depend on grouping and filtering; the appendix reports both nominal proportions and realized counts. Forward-only SFT uses all pairs in the forward direction. A replacement mixture partitions pairs by identity, keeping grouped variants of a program or country together, and replaces the selected examples with their reversals. No underlying pair is trained in both directions in a replacement mixture. Every mixture starts from the base model. Training-example counts, epochs, and optimizer steps match forward SFT; sequence lengths, supervised-token exposure, and runtime can differ and are reported separately.

Models, controls and uncertainty. We evaluate Llama 3B/8B, Gemma 4B/12B, and OLMo 1B, primarily with LoRA. The main reference comparison uses Llama 3B; additional models and training runs appear in Appendix B. Generic instruction replay replaces half of task examples, backward-only training tests backward learnability, and longer-forward and both-direction controls use additional budget. The full dose ladder and controls are in Table 9. A common effective batch and final-checkpoint recipe are detailed in Appendix C.1. Paired cluster-bootstrap intervals measure uncertainty over test examples; ranges across training runs separately show training variability. Neither repeated tasks nor resampled test examples count as independent task families.

Task	Input and required output	Primary success check	Train
Code transformation	Readable Python $\rightarrow$ obfuscated Python; e.g. renaming count to <code>a0</code> without changing execution. Backward requests readable code.	Execution agrees on stored tests and output is not a copy. Original identifier names are not required.	6499
Format conversion	JSON $\rightarrow$ YAML/XML; Markdown table $\rightarrow$ CSV. Backward converts back; e.g. <code>{"id": 3}</code> $\leftrightarrow$ <code>id: 3</code> .	Target parser accepts output, data structures agree, and output is not a copy. YAML syntax need not be exclusive to YAML.	6500
Translation	English $\leftrightarrow$ German/Chinese; both training orientations. E.g. "Good morning" $\leftrightarrow$ "Guten Morgen".	Target-language check and COMET-22 above a fixed model/direction threshold; backward additionally rejects copies.	6500
Question $\rightarrow$ SQL	Question plus schema $\rightarrow$ SQLite query; backward produces a question. E.g. "How many rows?" $\leftrightarrow$ <code>SELECT COUNT(*) FROM t</code> .	Forward: equal database results. Backward: a frozen parser maps the question to a query with equal results.	6499
Factual relations	Country $\rightarrow$ capital, ISO code or TLD; backward identifies the country. E.g. France $\leftrightarrow$ Paris.	Normalized first-line exact match, excluding copies. Ambiguous relation values are removed before splitting.	504
Execution prediction	Function and argument $\rightarrow$ return value; backward finds an argument. For $f(x) = x^2$, $3 \rightarrow 9$; backward accepts 3 or -3 .	Forward: return value agrees with execution. Backward: any argument executing to the requested value is correct.	302
Algebra	Factored $\rightarrow$ expanded polynomial, and the opposite training orientation. E.g. $(x+1)(x-1) \leftrightarrow x^2 - 1$.	Symbolic equivalence and no copy; factoring also requires a nonexpanded product. Expansion has no separate form check.	6500

Table 1: Task definitions and small illustrative examples (not scored test items). Train is the number of retained forward-SFT pairs for the Llama3B reference run. Sources: local Python transformation corpus, synthetic formats/algebra, News Commentary (training) and FLORES-200 (test), Spider, country-relation instruction datasets, and CRUXEval, respectively. Appendix A gives splits, exact prompt templates, thresholds, parsing and exclusions.

4 Forward gains can hide backward losses

Forward-only SFT substantially reduces backward performance in several tasks, sometimes while improving forward performance. Formatting gives the clearest example: forward success rises from 60.2% to 99.7%, while backward success falls from 51.1% to 0.4%. Code transformation shows a similar tradeoff: forward success rises from 37.5% to 82.4%, but backward success falls from 32.0% to 3.8%. Forward improvement alone would miss both losses.

Across the broader set of 91 evaluated model–task–training-run combinations, 52 meet the severe backward-loss threshold. Table 6 gives model coverage; these are descriptive counts, not pooled significance tests. Variation is substantive: German-to-English training also harms forward translation in the reference run, whereas factual-relation SFT improves both directions. Repeated unit-conversion experiments likewise improve both directions under the original recipe (Appendix C.6). Directional loss is therefore a property to measure, not a necessary outcome of fine-tuning.

5 Small backward doses with ordinary SFT

Small reversal proportions prevent much of the backward loss in formatting and code. Formatting backward success is 90.2% with 1% reversal and 96.2% with 5%, versus 0.4% under forward-only SFT. Code backward success is 55.8% at 1% versus 3.8%, while forward success changes from 82.4% to 82.8%. Generic replay and doubled forward training do not yield comparable backward benefits in these comparisons (Table 9). More forward optimization is not a substitute for supplying backward targets.

The benefits recur across 3 formatting runs and 3 code runs with different training seeds. At 1%, the backward improvement over forward SFT ranges from 86.1 to 89.8 percentage points in formatting and from 52.0 to 60.0 in code. These are within-run differences, not differences between independently evaluated checkpoints. Table 7 reports both-direction ranges for every task with repeated low-dose measurements; paired test uncertainty is reported separately.

Preservation beyond code and formatting. In English-to-Chinese training, backward translation success is 68.2% with 1% reversal versus 4.0%

Task	Dir.	Base	Forward SFT	1% reversal	5% reversal
German → English	F	73.7	67.1	67.0	67.4
	B	79.2	0.0	79.4	82.0
English → German	F	79.2	84.8	84.6	85.5
	B	73.3	64.4	70.8	68.5
English → Chinese	F	74.1	77.9	76.6	77.4
	B	75.4	4.0	68.2	68.3
Chinese → English	F	76.0	71.5	72.5	72.1
	B	74.0	0.0	76.2	77.1
Code transformation	F	37.5	82.4	82.8	84.3
	B	32.0	3.8	55.8	75.2
Question → SQL	F	49.5	62.4	62.3	60.8
	B	51.1	23.2	48.5	55.0
Format conversion	F	60.2	99.7	99.6	99.9
	B	51.1	0.4	90.2	96.2
Factual relations	F	71.9	86.7	—	—
	B	65.2	71.1	—	—
Polynomial expansion	F	17.3	26.1	25.5	26.8
	B	14.7	7.8	14.6	20.6
Polynomial factorization	F	14.6	43.6	44.0	43.8
	B	17.4	14.7	10.6	15.8
Execution prediction	F	26.5	29.0	—	—
	B	16.5	13.0	—	—

Table 2: Strict success (%); F/B denotes forward/backward. Blue highlights small reversed-pair doses; red highlights SFT backward success at most half the base rate (a descriptive threshold). Llama 3.2-3B, training seed 17, all eleven task orientations. Dose columns replace that share of forward pairs with their reversals, keeping instances and optimizer steps fixed. Preservation starts from the base checkpoint; this is not repair after forward-only training. Full doses and controls appear in Table 9, and variation across training runs in Table 7. Dashes denote proportions not yet evaluated. Relation/execution low doses are exploratory small-corpus additions.

with forward SFT. Forward success changes from 77.9% to 76.6%, revealing a cost. In question-to-SQL training, backward question success rises from 23.2% to 48.5% at 1% and 55.0% at 5%; forward success at 1% is 62.3% versus 62.4%. Figure 1 displays the joint forward/backward comparison; Appendix B separates paired uncertainty from variation across training runs.

The useful dose depends on the task. When training polynomial expansion, 1% reversed pairs bring backward factorization success to 14.6%, close to the base rate of 14.7%. The opposite training orientation behaves differently: when training factorization, 1% reversal lowers backward expansion success from 14.7% to 10.6%. At 5% it is 15.8%. Thus a dose that maintains one conditional can harm the other; the mixture must be judged jointly with forward performance and relative to the existing capability. The adverse 1% expansion result occurs in all 3 factorization-training runs. A uniform 1% rule would obscure this finding.

The full ladder evaluates 1, 5, 10, 25, and 50% replacement where available. Our prespecified dose knee is the smallest proportion reaching

90% of the half-reversal model’s backward success. This is a within-task criterion, not a common absolute target or a statistical equivalence test. Small relation/execution corpora yield fewer than one effective batch of distinct reversed examples at low doses; their later exploratory measurements are excluded from the original knee analysis.

6 What specialized objectives add

Specialized objectives do not consistently replace directional supervision, but some add benefits when combined with it. The comparison asks whether an objective improves generation beyond ordinary reversed-pair SFT, rather than whether it beats forward-only training alone. Table 3 reports both task directions and separate controls.

Information and objective controls. Our representation-alignment objectives combine forward-only or mixed-direction CE with symmetric contrastive alignment. Both sides are encoded independently and negatives come only from training data. The alignment term has no backward generation CE, but still exposes the inverse correspondence. Shuffled-positive

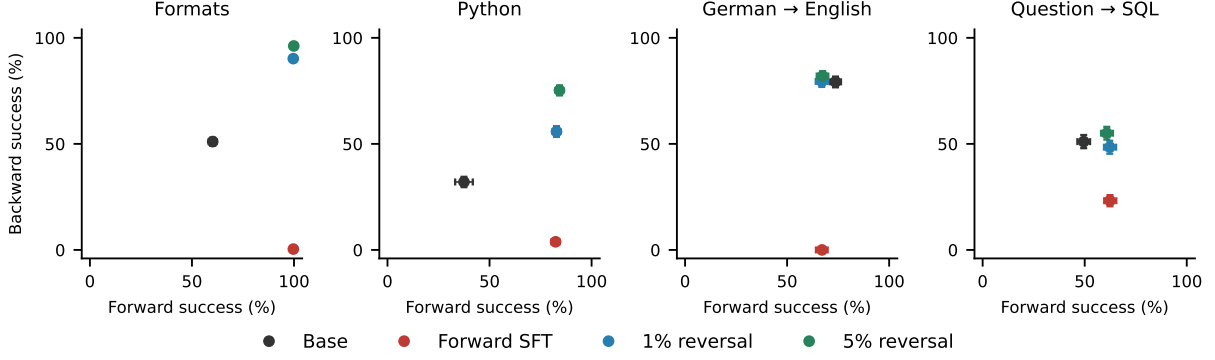


Figure 1: The preservation tradeoff for four tasks, comparing only the base model, forward-only SFT, and 1%/5% reversal. Each panel evaluates Llama3B on the same examples (training seed 17). Bars show marginal 95% cluster-bootstrap intervals over test examples. Table 8 gives paired differences; Table 7 shows variation across training runs. Point proximity does not establish equivalence.

(a) Contrastive objectives and exposure controls

Task	Model	Dir.	Base	SFT	5%	Replay	Rev.	CL-F	CL-M	Shuffle	CE-F+	CE-M+
German → English	G12B	F	76.9	66.9	66.5	67.3	1.6	66.1	66.8	66.1	66.5	66.6
		B	72.0	0.2	60.1	56.2	66.4	0.2	60.9	0.3	0.1	60.4
German → English	G4B	F	75.4	67.7	67.4	68.7	40.0	67.2	69.4	67.1	66.9	66.7
		B	75.8	0.0	74.4	70.4	76.8	0.0	74.5	0.0	0.0	75.1
German → English	L8B	F	74.5	64.2	65.8	63.9	4.7	62.8	64.5	63.8	62.8	65.1
		B	73.0	16.5	68.9	64.3	75.1	0.0	67.7	0.0	18.2	69.6
German → English	L3B	F	73.7	67.3	67.0	68.5	63.4	66.5	68.0	66.7	67.0	66.4
		B	79.7	0.0	82.0	61.0	84.9	0.0	82.9	0.1	0.0	82.2
Format conversion	G4B	F	78.3	100.0	99.9	99.8	52.2	100.0	99.9	99.9	99.9	100.0
		B	75.1	18.7	98.0	49.3	100.0	15.8	99.0	45.0	15.8	99.1
Format conversion	L8B	F	80.2	99.5	99.9	99.8	2.2	100.0	99.8	99.7	100.0	99.9
		B	69.0	0.0	98.7	32.1	99.9	2.2	99.2	0.0	0.0	98.7
Format conversion	L3B	F	60.6	99.7	99.9	98.9	27.0	99.6	99.6	99.7	99.2	99.7
		B	51.4	0.4	96.0	25.6	99.4	0.1	96.0	0.8	0.6	94.2
Unit conversion	G12B	F	33.7	97.1	97.0	94.0	44.1	97.6	96.5	97.5	97.4	97.1
		B	35.8	69.8	79.4	56.5	94.9	70.3	80.1	74.1	58.7	79.6

(b) Other auxiliary objectives

Task	Model	Dir.	Base	SFT	5%	CFT	Unlikelihood	Round-trip
English → German	L8B	F	71.1	73.1	73.0	<i>n/a</i>	74.0	—
		B	74.2	58.7	69.2	<i>n/a</i>	0.1	—
English → German	L3B	F	79.7	84.8	85.4	<i>n/a</i>	85.1	—
		B	72.8	63.6	68.8	<i>n/a</i>	0.0	—
Code transformation	L3B	F	38.0	81.4	84.2	84.1	—	—
		B	32.0	3.6	75.4	6.7	—	—
Question → SQL	L8B	F	62.0	72.1	71.9	<i>n/a</i>	—	71.3
		B	54.9	51.3	57.8	<i>n/a</i>	—	58.8
Question → SQL	L3B	F	49.1	62.3	61.0	<i>n/a</i>	—	61.5
		B	51.5	23.3	55.2	<i>n/a</i>	—	48.2

Table 3: Strict success (%); F/B means forward/backward. Colors follow Table 2. Contrastive and other auxiliary objectives, training seed 17. Each task/model comparison evaluates its own base and adaptations on the same examples. L/G denotes Llama/Gemma; B denotes billions of parameters. The 5% column is ordinary reversed-pair SFT. CL-F/M adds alignment to forward/mixed-direction CE; Shuffle randomizes positive pairs; CE-F+/M+ adds CE as an exposure proxy; CFT is the reconstructed code-equivalence contrastive fine-tuning recipe. Dashes denote objectives not yet evaluated; *n/a* marks code-equivalence CFT outside code. Paired contrasts and variation across training runs appear in Appendix B.

controls test pairing, and additional CE controls provide an exposure/computation proxy. They do not match measured FLOPs. CFT is a separate, code-specific reconstruction that trains program-equivalence judgments with CE; it is distinct from representation alignment. Other baselines add instruction-conflict unlikelihood or periodically generate a forward answer and train reverse round-trip CE. Their realized directional token exposure is reported separately.

Generation results. Forward-only representation alignment does not consistently improve backward generation beyond additional-CE or shuffled-pair controls. In code, CFT yields 6.7% backward success versus 75.4% for 5% reversed-pair SFT, with each evaluated on the same examples as its base and controls. Forward-only unlikelihood also fails to preserve backward translation in these experiments. These comparisons establish why directional SFT is a necessary baseline.

The objective comparison also has positive and mixed results. Mixed-direction alignment sometimes improves on the 5% mixture. In SQL/Llama8B, round-trip training yields 58.8% backward success versus 57.8% for the mixture, while forward success is 71.3% versus 71.9%. These point estimates trade a small forward decrease for a backward increase; their uncertainty is shown in Table 12. Appendix B reports all primary contrasts and their per-run uncertainty. Confidence intervals spanning zero do not show equivalence, especially near a generation floor.

7 Why backward examples may help

Forward CE trains y given forward instruction I_F and x , without directly penalizing an incorrect backward answer given I_B and y . Partition pairs into unreversed F_ρ and reversed B_ρ . With target-completion loss ℓ_θ , replacement has the schematic pair-level objective

$$\mathcal{L}_\rho = \frac{1}{|D|} \left[\sum_{(x,y) \in F_\rho} \ell_\theta(y \mid I_F, x) + \sum_{(x,y) \in B_\rho} \ell_\theta(x \mid I_B, y) \right]. \quad (1)$$

The relationship between the paired sides is unchanged, but the requested conditional changes. Our hypothesis is that backward examples maintain task-conditioned output selection that forward-only updates can bias. This explains why

ordinary CE could preserve a previously usable backward capability without a special alignment objective.

Supporting diagnostics examine this account, after the empirical preservation comparison. In formatting and translation, forward SFT reduces preference for correct backward answers over copied inputs; mixed-direction SFT increases that preference. Gemma translation is instructive: absolute loss on correct backward answers improves despite degraded generation, so answer likelihood alone does not explain the failure. Formatting middle-layer removals improve backward success beyond scaling and random removals matched for retained update norm, with small forward changes. This suggests selective interference in that task.

Final-checkpoint gradient cosines offer little evidence of persistent gradient conflict. Local backward updates improve held-out loss even in the units null case, so such an improvement is not a collapse-specific signature. Candidate lengths, valid alternative translations, unmatched edit norms and multiple interventions limit interpretation. Appendix D gives the complete diagnostic comparisons. These results support an output-selection hypothesis without demonstrating erased knowledge, a universal cause, or a general post-training repair.

8 Related work

Directional degradation and reverse training. Code specialization (Nikiema et al., 2025), direction-dependent translation loss (Zhu et al., 2024), and program-execution asymmetry (Hasanov et al., 2026) motivate evaluating existing abilities in both directions. The reversal curse concerns a reverse capability that was never acquired (Berglund et al., 2024). Reverse training addresses direction at the data level (Golovneva et al., 2024); we measure preservation across doses and compare objectives against that supervision. Bidirectional behavior need not imply a unified representation (Coda-Forno et al., 2026).

Forgetting, rehearsal and recovery. Forgetting after fine-tuning (Luo et al., 2023), differences between LoRA and full fine-tuning (Biderman et al., 2024), and rehearsal (Scialom et al., 2022) motivate our controls. Wrapper and offset accounts (Jain et al., 2024; Lee et al., 2024), skewed task inference (Kotha et al., 2024), and update filtering (Abro and Tahir, 2026) concern distinct explana-

tions or later repair. Our replacement intervention preserves during training; behavioral repair alone does not identify a unique internal mechanism.

Objective attribution. Data mixtures can confound attribution (Longpre et al., 2023; Lipton and Steinhardt, 2018). RevThink adds backward questions and reasoning (Chen et al., 2025), illustrating why directional exposure matters even for a bidirectional objective. Reversed-pair SFT, additional CE, and shuffled positives test what an objective adds without assuming auxiliary objectives are universally unnecessary.

9 Conclusion

Training a model in one direction can damage an existing reverse-task ability. In several tested settings, replacing a small fraction of forward examples with reversals preserves much of backward performance while retaining most forward performance. The useful dose and forward cost vary across tasks; some specialized objectives add further benefits. The practical implication is to evaluate both directions and compare new preservation objectives against ordinary directional supervision. Output-selection diagnostics help interpret this result and remain a supporting hypothesis.

Limitations

Scoring and alternatives. Correctness is task-specific. Code execution on finite tests does not prove semantic equivalence or readable-source recovery. Format equality normalizes scalar case and does not enforce YAML-exclusive syntax. Algebra expansion lacks a separate expanded-form check, and factorization need not be complete. Factual exact matching may reject aliases. Translation thresholds differ by model/direction; its original forward predicate lacks explicit echo rejection. SQL question correctness depends on a frozen parser that can reject valid paraphrases. The proposed WebNLG text check fails on reference texts and therefore cannot support a tuned comparison (Appendix A).

Scope and generality. Replacement cannot recover information absent from the input. Original code is Python-only; translation covers English paired with German or Chinese. Synthetic variants and changed output contracts reuse semantic corpora rather than supplying independent domains.

Some proposed tasks have inadequate starting bidirectional performance or unreliable checks; Appendix C.6 reports these exclusions. Independent OPUS transfer fails its copy-validity criterion, so it establishes no tuned preservation result.

Models, training and resolution. Coverage is uneven across five model sizes and training runs, and most comparisons use LoRA. Full-weight checks cover selected Llama3B runs and do not establish a bound on other recipes. Example and optimizer-step matching does not imply matched tokens or runtime. Paired intervals quantify test uncertainty; training-run ranges do not establish an optimal dose or equivalence between adjacent proportions. Changed prompts and recipes can introduce forward costs or backward loss, including in the original units null case. Historical upstream dataset revisions and training versions are incomplete, limiting exact reconstruction beyond the retained corpora and result analysis.

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516	A Task data, prompts and correctness			
517	Strict success is a binary task-specific predicate,			
518	not a common metric across tasks. The pri-			
519	mary evaluation uses greedy decoding and the sim-			
520	ple instruction without demonstrations. A shared			
521	system message and the same backward instruc-			
522	tion are used in training and evaluation. Output			
523	extraction strips complete or opened Markdown			
	code fences. Echo detection compares Unicode-			
	NFKC/whitespace-normalized strings and also re-			
	jects character-trigram Jaccard overlap at least			
	0.95 when both strings have at least twelve char-			
	acters. Domain-specific exceptions are stated be-			
	low; no new scoring rule is applied to the reported			
	results.			
	Dataset versions and splits. The saved paired			
	JSONL corpora, resolved build configurations,			
	prompt/scorer sources and SHA-256 hashes are			
	listed in the accompanying task-methods metadata.			
	Some original upstream dataset revisions were not			
	recorded; current cached Hub revisions cannot es-			
	tablish the historical training version. The frozen			
	paired corpora therefore specify the reproducible			
	analysis inputs, without implying exact reconstruc-			
	tion of all historical training data.			
	Shared system message.			
	You are a precise transformation tool. You con-			
	vert the given input into the requested form ex-			
	actly as instructed, preserving its meaning, and			
	you never change what it says or computes. Re-			
	ply with the result only: no explanation, no com-			
	mentary, and no markdown code fences.			
	A.1 Python transformation			
	The local Python corpus supplies read-			
	able/obfuscated pairs over five transformations:			
	misleading identifier renaming (L1b), random			
	hexadecimal renaming (L1r), sequential minifica-			
	tion (L2), control-flow flattening (S1), and opaque			
	predicates/dead code (S2). Training/validation			
	use the transformation-generation pool; held-out			
	tests include only programs available under all			
	five transformations. A program is the uncertainty			
	cluster. Renaming destroys the original names, so			
	the inverse is not unique. Both primary directions			
	require agreement on every stored execution test			
	and reject copies. Transformation-style compli-			
	ance, original-name recovery, CodeBLEU and			
	readability are not conjuncts of primary success;			
	the workshop readability/similarity predicate			
	is secondary. Built train/validation/test pairs:			
	6500/500/2060. Retained reference forward-SFT			
	pairs: 6499. Realized reference doses: 1%: 62			
	reversed pairs among 6499 retained pairs; 5%:			
	326 reversed pairs among 6499 retained pairs.			
	Forward prompt.			
	Rewrite the following Python program			
	so that it is obfuscated in the same			
	style as the transformation named below,			
	without changing what it computes.			

575	Transformation: <TRANSFORMATION>		
576	Program:		
577	<INPUT A>		
578	Reverse prompt.		
579	Recover the original, readable Python		
580	source that this obfuscated program was		
581	produced from.		
582	Program:		
583	<INPUT B>		
584	A.2 Serialization conversion		
585	A seeded generator creates lossless JSON/YAML,		
586	JSON/XML, and Markdown-table/CSV pairs.		
587	JSON uses json.loads, YAML yaml.safe_load,		
588	XML an ElementTree decoder, CSV csv.reader,		
589	and Markdown a pipe-table parser that removes		
590	separator rows. Equality requires matching dictio-		
591	nary keys, container shapes and list order; scalar		
592	strings are stripped and lowercased to accommo-		
593	date XML/CSV type conversion. Copies are re-		
594	jected in both directions. JSON can parse as		
595	YAML: the copy guard excludes unchanged or		
596	near-unchanged inputs, but this scorer does not		
597	enforce YAML-exclusive syntax or case-sensitive		
598	scalar identity. Built train/validation/test pairs:		
599	6500/500/2500. Retained reference forward-SFT		
600	pairs: 6500. Realized reference doses: 1%: 65 re-		
601	versed pairs among 6500 retained pairs; 5%: 325		
602	reversed pairs among 6500 retained pairs.		
603	Forward / json-yaml prompt.		
604	Convert the following JSON document into		
605	YAML, preserving every key and value.		
606	JSON:		
607	<INPUT A>		
608	Reverse / json-yaml prompt.		
609	Convert the following YAML document into		
610	JSON, preserving every key and value.		
611	YAML:		
612	<INPUT B>		
613	Forward / json-xml prompt.		
614	Convert the following JSON document into		
615	XML, preserving every key and value.		
616	JSON:		
617	<INPUT A>		
618	Reverse / json-xml prompt.		
619	Convert the following XML document into		
620	JSON, preserving every key and value.		
621	XML:		
622	<INPUT B>		
	Forward / mdtable-csv prompt.		623
	Convert the following Markdown table		624
	document into CSV, preserving every key		625
	and value.		626
	Markdown table:		627
	<INPUT A>		628
	Reverse / mdtable-csv prompt.		629
	Convert the following CSV document into		630
	Markdown table, preserving every key and		631
	value.		632
	CSV:		633
	<INPUT B>		634
	A.3 Machine translation		635
	Training uses the Helsinki-		636
	NLP/news_commentary Hub export, German-		637
	English and English-Chinese configurations.		638
	Both sides must have 20–400 characters;		639
	normalized source duplicates are removed.		640
	Test data use FLORES-200 devtest from the		641
	haoranxu/FLORES-200 mirror. Each orientation		642
	trains its designated source-to-target mapping.		643
	COMET uses Unbabel/wmt22-comet-da with		644
	source, generated translation and reference.		645
	Success requires nonempty output, the target-		646
	language check, and COMET at or above the		647
	frozen threshold. Backward additionally rejects		648
	copies; the original forward predicate does not.		649
	Chinese requires a Han-character fraction above		650
	0.15; English/German use langdetect probability		651
	above 0.5 and reject Han characters. A language-		652
	detector exception accepts the language check, a		653
	limitation. chrF++ and BLEU are secondary. The		654
	threshold is the 25th percentile of scores from		655
	200 untouched-model outputs for each model and		656
	direction, before tuned evaluation. Thus success		657
	percentages compare training methods within a		658
	model/direction, not absolute translation quality		659
	across models. Built train/validation/test pairs:		660
	6500/500/1012. Retained reference forward-SFT		661
	pairs: 6500. Realized reference doses: 1%: 65		662
	reversed pairs among 6500 retained pairs; 5%:		663
	325 reversed pairs among 6500 retained pairs.		664
	Forward prompt.		665
	Translate the following English text		666
	into German.		667
	English:		668
	<INPUT A>		669

670	Reverse prompt.		
671	Translate the following German text into		
672	English.		
673	German:		
674	<INPUT B>		
675	A.4 Semantic parsing and verbalization		
676	Questions and queries come from the official		
677	database-disjoint xlangai/spider train/validation		
678	splits; SQLite databases come from the prem-		
679	research/spider mirror, whose pooled question		
680	files are not used. The official training split		
681	is shuffled into training and validation; official		
682	development databases form the test set. Both		
683	prompts include a schema summary. Forward		
684	queries are checked by the Spider test-suite ex-		
685	ecution evaluator on the example database, with		
686	plug_value=False and keep_distinct=False; alter-		
687	natives with equal denotations can pass. Reverse		
688	questions are parsed by frozen ibm-granite/granite-		
689	3.1-8b-instruct and checked by the same execu-		
690	tion comparison. Empty outputs, copied queries		
691	and outputs containing SELECT are rejected in re-		
692	verse. Parser errors can reject a valid paraphrase.		
693	Its reference-question accuracy is a calibration di-		
694	agnostic, not a mathematical ceiling for every gen-		
695	erated paraphrase. Built train/validation/test pairs:		
696	6499/500/1034. Retained reference forward-SFT		
697	pairs: 6499. Realized reference doses: 1%: 65 re-		
698	versed pairs among 6499 retained pairs; 5%: 325		
699	reversed pairs among 6499 retained pairs.		
700	Forward prompt.		
701	Database schema:		
702	<DATABASE SCHEMA>		
703	Write one SQLite query that answers this		
704	question.		
705	Question:		
706	<INPUT A>		
707	Reverse prompt.		
708	Database schema:		
709	<DATABASE SCHEMA>		
710	Write the question, in plain English,		
711	that this SQLite query answers.		
712	Query:		
713	<INPUT B>		
	A.5 Country relations		714
	Lots-of-LoRAs/task1146_country_capital,		715
	task1314_country_abbreviation and		716
	task1320_country_domain_tld provide country-		717
	capital, country-ISO-code and country-TLD		718
	pairs. Values associated with multiple countries		719
	and identical input/target strings are removed;		720
	many-to-one currency/calling-code relations		721
	are excluded. Splits and uncertainty clusters		722
	are countries, keeping their relations together.		723
	The scorer uses the first output line, removes a		724
	leading "the" or "answer" marker and a trailing		725
	period, then compares after Unicode/whitespace		726
	normalization. It is case-sensitive and does not		727
	accept unrestricted aliases or alternative names.		728
	Copying is rejected. Built train/validation/test		729
	pairs: 504/69/135. Retained reference forward-		730
	SFT pairs: 504. Realized reference doses: 1%: 6		731
	reversed pairs among 504 retained pairs; 5%: 20		732
	reversed pairs among 504 retained pairs.		733
	Forward / capital prompt.		734
	Name the capital city of the following.		735
	Answer with the name only.		736
	<INPUT A>		737
	Reverse / capital prompt.		738
	Name the country whose capital is the		739
	following. Answer with the name only.		740
	<INPUT B>		741
	Forward / abbrev prompt.		742
	Name the two-letter ISO country code for		743
	the following. Answer with the name		744
	only.		745
	<INPUT A>		746
	Reverse / abbrev prompt.		747
	Name the country whose two-letter ISO		748
	code is the following. Answer with the		749
	name only.		750
	<INPUT B>		751
	Forward / tld prompt.		752
	Name the internet top-level domain of		753
	the following. Answer with the name		754
	only.		755
	<INPUT A>		756
	Reverse / tld prompt.		757
	Name the country whose internet		758
	top-level domain is the following.		759
	Answer with the name only.		760
	<INPUT B>		761

A.6 Output and input prediction

The cruxeval-org/cruxeval test export contains 800 Python function/input/output problems. Corpus construction removes identical input/output strings, unexecutable programs, inconsistent reference outputs, and problems where feeding the output back into the function is a valid reverse answer. The retained corpus is split by program. Forward compares the predicted literal with the canonicalized executed return value. Reverse executes the generated argument and accepts any preimage producing the required return value; matching the stored input is unnecessary. Code fences are stripped. Forward answers and reference return values are parsed with `ast.literal_eval`, falling back to stripped text; generated reverse arguments are passed directly to the execution harness. Execution-based canonicalization determines equality. Built train/validation/test pairs: 302/100/200. Retained reference forward-SFT pairs: 302.

Forward prompt.

Given this Python function, what does it return for the input shown?

```
<FUNCTION CODE>
```

Call:

```
f(<INPUT A>)
```

Write only the returned value, as a Python literal.

Reverse prompt.

Given this Python function, find an argument that makes it return the value shown.

```
<FUNCTION CODE>
```

Required return value:

```
<INPUT B>
```

Write only the argument, as a Python literal.

A.7 Expansion and factorization

A seeded SymPy generator multiplies primitive integer-coefficient linear factors into degree-2–4 polynomials in one or two variables, using the configured root range. Expanded symbolic expressions are deduplicated across splits. The expansion orientation has factored inputs and expanded targets; the factorization orientation swaps the sides of the same generated pairs. Answers allow caret notation and implicit multiplication.

Success requires symbolic equivalence after expansion and excludes copies. Factoring also requires a non-Add top level whose expansion differs from itself; it does not require a canonical or complete irreducible factorization. The expansion scorer has no separate expanded-form conjunct, despite the prompt asking for a polynomial. Built train/validation/test pairs: 6500/500/2500. Retained reference forward-SFT pairs: 6500. Realized reference doses: 1%: 65 reversed pairs among 6500 retained pairs; 5%: 325 reversed pairs among 6500 retained pairs.

Forward prompt.

Expand this expression and collect like terms. Write the result as a polynomial.

Expression:

```
<INPUT A>
```

Reverse prompt.

Factor this polynomial completely over the integers. Write it as a product of factors.

Polynomial:

```
<INPUT B>
```

A.8 Other translation and algebra orientations

The following templates complete the opposite training directions and Chinese pair. Placeholders identify the input side in that orientation; the shared system message is unchanged.

German → English / forward prompt.

Translate the following German text into English.

German:

```
<INPUT A>
```

German → English / reverse prompt.

Translate the following English text into German.

English:

```
<INPUT B>
```

English → Chinese / forward prompt.

Translate the following English text into Chinese.

English:

```
<INPUT A>
```

English → Chinese / reverse prompt.

Translate the following Chinese text into English.

Chinese:

```
<INPUT B>
```

Chinese → English / forward prompt.

Translate the following Chinese text into English.

Chinese:

<INPUT A>

Chinese → English / reverse prompt.

Translate the following English text into Chinese.

English:

<INPUT B>

Polynomial factorization / forward prompt.

Factor this polynomial completely over the integers. Write it as a product of factors.

Polynomial:

<INPUT A>

Polynomial factorization / reverse prompt.

Expand this expression and collect like terms. Write the result as a polynomial.

Expression:

<INPUT B>

A.9 Translation thresholds

Thresholds below are frozen separately by model, orientation and direction; each row uses the base model calibration described above. They must not be used to rank models by absolute translation quality.

A.10 Model-based correctness checks

SQL-to-question is the model-based check in the main comparison: a frozen Granite parser maps a generated question back to SQL, whose database result is compared with the target. The other proposed round-trip direction is WebNLG triples-to-text: the same model family extracts triples from generated text. Its extractor fails on reference texts, so this task supplies no tuned-model result and is absent from the main task table. Neither check is executable or symbolic verification of the generated natural-language meaning itself.

B Complete model coverage and preservation comparisons

The main comparison displays one reference training run, while the tables below retain variation across runs, paired test uncertainty, the full dose ladder and objective controls. The severe-loss highlighting is descriptive rather than a statistical

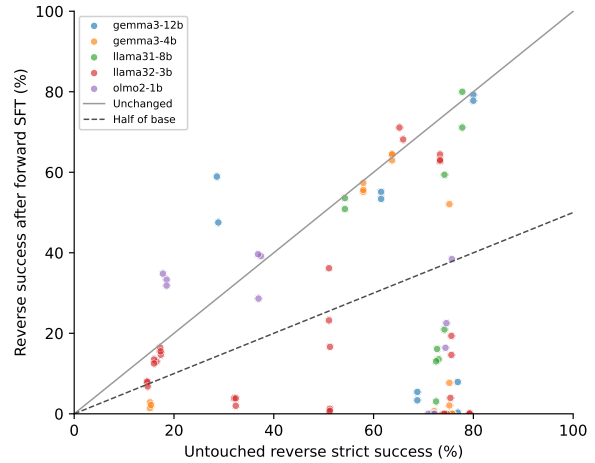


Figure 2: Base and forward-SFT backward success over model–task–training–run combinations. The dashed line marks half the base rate. Repeated tasks and training runs are not independent observations.

significance claim. Repeated evaluations use the latest complete result, chosen without inspecting the effect sign; earlier results remain in the accompanying data.

C Additional methods, experiments and boundary conditions

These experiments test forward retention, general ability, sensitivity to prompts and training recipes, and the scope of preservation on other tasks. Exclusions and revised criteria are reported with their scientific reasons. Test outcomes were not used to choose a favorable training run or diagnostic subset.

C.1 Paired data and reference training recipe

Each task supplies fixed training, validation, and test pairs. Content and pair-identity overlap checks exclude held-out test material during corpus construction; validation is not used to select a preferred test result. Opposite orientations of a task share the underlying task family. The reference recipe and actual corpus/trainer counts are reported in Table 14; smaller factual-relation and execution corpora are retained at their actual size. Seed-specific partitions define reversed-pair replacement. Training summaries retain realized supervised-token exposure and length filtering, rather than equating pair count with tokens. Every reported tuned comparison includes a freshly evaluated base and all declared arms on the same retained instance set.

Task orientation	Model	Forward τ	Backward τ
English $\rightarrow$ German	L3B	0.764888	0.861910
English $\rightarrow$ German	L8B	0.839859	0.871522
English $\rightarrow$ German	G4B	0.836251	0.869601
English $\rightarrow$ German	G12B	0.866418	0.874782
English $\rightarrow$ German	O1B	0.555442	0.779690
German $\rightarrow$ English	L3B	0.860885	0.764888
German $\rightarrow$ English	L8B	0.871522	0.836640
German $\rightarrow$ English	G4B	0.869446	0.832958
German $\rightarrow$ English	G12B	0.874782	0.866888
German $\rightarrow$ English	O1B	0.779690	0.555477
English $\rightarrow$ Chinese	L3B	0.787774	0.823006
English $\rightarrow$ Chinese	L8B	0.843215	0.844090
English $\rightarrow$ Chinese	G4B	0.850115	0.844765
English $\rightarrow$ Chinese	G12B	0.863640	0.859140
English $\rightarrow$ Chinese	O1B	0.636863	0.696096
Chinese $\rightarrow$ English	L3B	0.822380	0.787968
Chinese $\rightarrow$ English	L8B	0.844090	0.844345
Chinese $\rightarrow$ English	G4B	0.845060	0.850115
Chinese $\rightarrow$ English	G12B	0.859140	0.865201
Chinese $\rightarrow$ English	O1B	0.698151	0.625003

Table 4: Frozen COMET-22 strict-success thresholds.

Model	n	Reference accuracy
G12B	200	63.0%
G4B	200	63.0%
L8B	200	62.5%
L3B	200	64.5%
O1B	200	62.5%

Table 5: Frozen SQL parser calibration on reference questions in the corresponding initial evaluation subset. Variation reflects the subsets and execution layout, not a parser tuned to each evaluated model.

Recorded reference recipe. The main-table Llama3B experiments use the common training recipe: LoRA rank 32, scaling 64, dropout 0.05, learning rate 0.0001, 3 epochs, effective batch 64, and maximum sequence length 2048 tokens. The scheduler is cosine with warm-up fraction 0.03, weight decay 0 and gradient clipping 1. Training uses bfloat16 and final checkpoints without intermediate checkpoint selection. LoRA targets query, key, value and output attention projections and the gate, up and down MLP projections. Allocation-specific microbatches and accumulation retain the recorded effective batch. Full-weight and robustness variants are separate recipes, not substitutions for this reference.

C.2 Dose, forward retention, and computational cost

The main figure plots forward and backward strict success for the base model, forward SFT, and the two smallest reversed-pair doses. The full lad-

der and secondary controls are reported separately. Panels retain model, task orientation, and seed identity. Arms lacking a common evaluation campaign are not connected as paired observations. The units boundary case is shown alongside collapse cells. Repeated passes use the latest complete evaluation, with earlier passes retained in the audit; selection never uses effect sign.

Paired contrasts use the domain’s instance/cluster unit. Training-seed variation is separate: resampling test instances does not measure training variability. No aggregate pools cells as independent domains. The dose knee uses the criterion in §5; overlapping intervals do not establish equivalence.

The cost table reports available trainer runtime, optimizer steps, and CE-corpus tokens. Auxiliary encoded-side exposure is retained separately in the analysis. Missing timing, GPU-type, queue-time, or exposure records remain unavailable. Additional CE is the declared proxy; corpus token counts alone do not establish matched objective FLOPs or wall time.

Scoped finding. On the representative translation campaign, reversed-pair SFT preserves reverse behavior with nearly unchanged forward task performance relative to forward SFT (Table 15). Generic replay recovers less reverse performance in that campaign. The units case improves both ways under ordinary SFT; robustness results show that preservation and forward costs depend on task and recipe. Recorded budgets sup-

Model	Task orientations	Training runs	Severe losses
gemma3-12b	7	14	8
gemma3-4b	7	21	14
llama31-8b	6	12	7
llama32-3b	11	32	18
olmo2-1b	4	12	5

Table 6: Evaluated model–task–training–run combinations. Severe loss means SFT backward success is at most half the base rate on the same examples. Counts are descriptive and are not independent task counts or pooled significance tests.

Task	Runs	ΔF_{1-S}	ΔB_{1-S}	ΔF_{5-S}	ΔB_{5-S}
German $\rightarrow$ English	3	[-1.0, +0.1]	[+76.1, +79.6]	[-0.6, +0.3]	[+80.0, +82.0]
English $\rightarrow$ German	3	[-0.9, -0.2]	[+6.2, +8.6]	[-1.6, +0.7]	[+4.1, +5.8]
English $\rightarrow$ Chinese	3	[-1.7, -0.8]	[+49.5, +64.2]	[-1.5, -0.5]	[+49.1, +64.3]
Chinese $\rightarrow$ English	3	[-0.7, +1.5]	[+76.2, +77.5]	[-0.1, +0.6]	[+77.1, +77.8]
Code transformation	3	[-0.2, +1.7]	[+52.0, +60.0]	[+0.0, +1.9]	[+68.8, +71.4]
Question $\rightarrow$ SQL	3	[-0.5, +2.0]	[+13.9, +33.4]	[-1.6, -0.5]	[+19.3, +40.0]
Format conversion	3	[-0.5, -0.0]	[+86.1, +89.8]	[-0.2, +0.2]	[+92.2, +95.8]
Polynomial expansion	3	[-1.6, -0.6]	[+6.8, +8.3]	[-2.2, +0.7]	[+12.8, +14.0]
Polynomial factorization	3	[-3.4, +0.4]	[-4.1, -3.4]	[-2.3, +0.1]	[-0.1, +1.1]

Table 7: Training-run variation for every Llama3B task orientation with at least two runs measuring both 1% and 5% reversal. Entries are minimum/maximum within-run differences from forward SFT, in percentage points; they are not confidence intervals. Every available qualifying run is included regardless of outcome. F/B means forward/backward; subscripts 1 and 5 identify reversal proportions. Other tasks and model coverage remain in Table 6.

port within-recipe comparisons, not matched wall-time claims across different objectives.

C.3 General ability and overlap controls

The fixed panel is German-to-English/Llama3B, format conversion/Gemma4B, and units/Gemma12B. Each campaign compares base, SFT, replay, mixed-direction SFT, forward-only and mixed-direction contrastive arms, and their additional-CE controls. The auxiliary arm is fixed rather than selected by its best test score.

IFEval measures instruction following without demonstrations; GSM8K measures numerical reasoning using fixed training demonstrations. A neutral assistant system prompt is shared by all systems. Dataset revisions and overlap exclusions are recorded before generation. Auditing covers both sides of every local training/validation corpus, replay, and mixed-task sources, using normalized exact/contained strings. Fixed demonstrations are checked against the retained test set. Lexical auditing does not establish semantic or pretraining independence; shared numerical structure limits interpretation of the units/GSM8K comparison.

All systems were evaluated on the same examples; generated-output differences from the base were checked to detect an adapter that failed to

load. Strict IFEval and strict GSM8K endpoints are reported separately; loose instruction checks and flexible numeric extraction remain secondary. Endpoint changes are paired within each campaign and seed variation is retained. A concentration claim is scoped to this declared panel; unrelated benchmark scales are not subtracted as if interchangeable.

Scoped finding. General retention depends on the endpoint and task (Table 17). Formatting/Gemma4B loses numerical reasoning under forward SFT while instruction-following changes are small; replay recovers more of that reasoning endpoint than mixed-direction SFT. Translation/Llama3B loses instruction following while improving numerical reasoning. Reverse-task recovery therefore does not establish general-ability preservation, and the evidence does not support a universal disproportionate-loss claim across benchmarks.

C.4 Independent real-data transfer

Existing German-to-English checkpoints are evaluated on the independently sourced OPUS-100 German–English test split.¹ The panel includes

¹<https://huggingface.co/datasets/Helsinki-NLP/opus-100>.

Task	Mixture	Forward change	Backward change
Format conversion	1%	-0.0 [-0.3, +0.2]	+89.8 [+88.6, +90.9]
Format conversion	5%	+0.2 [-0.0, +0.4]	+95.8 [+95.0, +96.5]
Code transformation	1%	+0.4 [-0.9, +1.7]	+52.0 [+49.4, +54.5]
Code transformation	5%	+1.9 [+0.5, +3.3]	+71.4 [+68.9, +73.8]
German → English	1%	-0.1 [-1.6, +1.5]	+79.4 [+76.9, +81.9]
German → English	5%	+0.3 [-1.4, +2.0]	+82.0 [+79.6, +84.3]
Question → SQL	1%	-0.1 [-1.5, +1.3]	+25.2 [+22.1, +28.3]
Question → SQL	5%	-1.5 [-3.2, +0.0]	+31.8 [+28.5, +35.0]

Table 8: Paired reversal-mixture changes from forward SFT for Figure 1, in percentage points with 95% cluster-bootstrap intervals (2,000 draws). These exploratory intervals are not adjusted for multiple comparisons and measure test-instance uncertainty, not training-run variability.

Llama3B and Gemma4B across the fixed training runs and the same SFT, replay, mixed-direction, contrastive, and additional-CE systems. Dataset revision, character-length inclusion rules, directions, overlap exclusions, and scorer are frozen before generations. A new prompt on an existing semantic corpus remains a diagnostic rather than independent transfer evidence.

Overlap auditing includes all local training, validation, replay, and original test sources. Gold, echo, empty, and invalid-output oracles run on compute. Original model-specific COMET thresholds are retained. Strict success, continuous COMET, chrF++, echo, and off-target rates remain separate; checkpoint choice, decoding, and scoring are not tuned on this corpus.

The base model is evaluated first. A task comparison requires initial success of at least 10% in each direction, format failures at most 15%, and copied-input strict success at most 2%. These requirements separate an existing bidirectional ability from a scoring failure or a capability that the model did not possess.

Completed measurements. On every transfer subset, copying the input exceeds the permitted strict-success rate. This violates the scorer-validity requirement and excludes tuned-model comparisons. The base-only boundary trials are retained in Table 18; no tuned checkpoint comparison is eligible. These measurements therefore do not establish preservation on this independent corpus, and the failed criterion is not evidence of adapter collapse.

Versioned diagnostics. Revised output contracts were specified after inspecting unsuccessful task/scorer checks. The original exclusions remain reported separately. Separate versions test bounded signed C++ literal syntax, explicit serial-

ization/conversion/logic contracts, and a WebNLG extractor supplied with a global training-only predicate schema. Translation transfer additionally excludes source copies in both directions, retaining the original thresholds. Reference-answer, copied-input and invalid-answer tests check the revised scorers before comparing tuned models; the base model must have adequate performance in both directions. Revised criteria and prompts define new cells; their results do not replace original failures or support cross-version paired contrasts. Failed oracles and genuine capability limits remain reported boundaries.

Completed versioned-domain pilots. Table 19 reports every completed pilot under the revised contracts, including all training runs and both directions. Formatting/Llama8B and units/Gemma12B improve backward performance under forward-only SFT; their mixed-direction gains are additional backward learning rather than restoration of a collapsed capability. Typed Python/C++ at Gemma12B recovers backward success with mixed-direction SFT, with seed-dependent forward costs. At Llama8B, the mixture fails to recover backward behavior consistently, whereas generic replay retains it much better; backward-only training also fails. These results bound the remedy and separate backward learnability from directional preservation. This panel tests only the half-reversal mixture, so it supplies no evidence for a small-dose optimum. Boolean tasks still lack adequate initial bidirectional performance. The WebNLG extractor fails its reference-text oracle, so no D2T tuned comparison is eligible.

C.5 Prompt and training-recipe robustness

The prompt panel includes translation/Llama3B, format conversion/Gemma4B, units/Gemma12B,

Task	Dir.	Reference		Reversed-pair share					Other adaptations				
		Base	SFT	1%	5%	10%	25%	50%	Replay	Rev.	Fwd2×	Flip	Multi
German → English	F	73.7	67.1	67.0	67.4	66.2	68.6	67.4	68.6	63.5	57.8	64.4	69.1
	B	79.2	0.0	79.4	82.0	82.8	82.3	82.9	59.0	85.2	1.4	83.9	71.5
English → German	F	79.2	84.8	84.6	85.5	85.4	84.7	84.6	85.4	0.0	77.8	84.8	84.0
	B	73.3	64.4	70.8	68.5	68.2	68.2	67.8	72.0	66.2	64.4	67.0	73.0
English → Chinese	F	74.1	77.9	76.6	77.4	77.1	78.6	76.7	78.6	0.0	75.9	80.4	77.0
	B	75.4	4.0	68.2	68.3	69.9	71.5	71.9	70.9	71.2	9.5	72.8	70.3
Chinese → English	F	76.0	71.5	72.5	72.1	72.3	72.2	71.8	74.8	23.4	63.8	71.5	73.3
	B	74.0	0.0	76.2	77.1	80.0	78.3	77.2	1.0	77.7	0.0	76.1	71.0
Code transformation	F	37.5	82.4	82.8	84.3	83.6	84.4	83.9	74.8	65.6	88.3	86.3	70.5
	B	32.0	3.8	55.8	75.2	76.5	80.3	85.6	14.5	86.3	9.6	86.6	3.4
Question → SQL	F	49.5	62.4	62.3	60.8	62.0	60.1	58.6	59.8	42.0	61.1	60.8	56.8
	B	51.1	23.2	48.5	55.0	57.5	59.9	58.1	17.1	59.1	27.3	58.8	50.4
Format conversion	F	60.2	99.7	99.6	99.9	99.8	99.8	99.8	99.0	25.2	99.0	99.8	97.8
	B	51.1	0.4	90.2	96.2	97.0	98.6	99.6	24.6	99.4	0.7	99.6	42.9
Factual relations	F	71.9	86.7	–	–	–	86.7	87.4	88.1	79.3	89.6	89.6	83.0
	B	65.2	71.1	–	–	–	80.0	80.7	67.4	83.7	58.5	81.5	65.9
Polynomial expansion	F	17.3	26.1	25.5	26.8	27.3	25.5	24.8	25.8	14.6	29.6	27.7	21.2
	B	14.7	7.8	14.6	20.6	25.3	31.0	36.5	9.2	43.6	6.7	44.9	6.0
Polynomial factorization	F	14.6	43.6	44.0	43.8	43.2	40.2	36.1	36.9	7.8	49.6	44.7	26.8
	B	17.4	14.7	10.6	15.8	18.0	23.0	24.9	14.0	26.2	12.7	27.4	26.5
Execution prediction	F	26.5	29.0	–	–	–	29.5	29.5	27.0	21.0	31.5	30.5	24.5
	B	16.5	13.0	–	–	–	26.0	28.0	13.0	30.0	12.5	29.0	12.5

Table 9: Full dose ladder and secondary controls, strict success (%). Llama 3.2-3B, training seed 17. F/B means forward/backward. Replay replaces half of task examples with generic instructions; Rev. trains backward only; Fwd2× doubles forward training; Flip trains both directions at doubled budget; Multi mixes other tasks. Each task uses the base model evaluated on the same examples as every adaptation. Dashes denote proportions not yet evaluated. Low-dose relation/execution experiments have fewer than one effective batch of distinct reversed pairs; these exploratory additions are excluded from the original dose-knee analysis.

and Python-C++/Llama8B. Frozen variants retain the original primary instruction and add fixed task/contract prefixes. Both directions use the same semantic instances, decoding limits, and scorers. Every frozen prompt is retained, separating echo/off-target output, semantic error, and measurement failure. Best-prompt summaries remain exploratory recovery diagnostics.

Synthetic extrapolation additionally uses exact unit magnitudes outside original support and behavior-disjoint piecewise-quadratic Python/C++ functions. Original allowlisted scorers and input ranges remain unchanged; reference outputs are checked on all generated instances. These tests do not establish unrestricted natural-code, loop/list, or independent-domain generality.

The recipe panel fixes lower/higher learning-rate and LoRA-rank variants around the original recipe, varying the factors separately. Parameter values, cells, seeds, and primary contrasts are fixed in the recorded protocol. The comparisons preserve alpha/rank ratio, effective batch, paired-data identity, direction mixture, sequence limit, epoch count, and optimizer schedule. The

audit records realized exposure, step count, and available GPU cost. SFT and mixed-direction SFT are compared on format conversion/Llama3B and units/Gemma12B, retaining collapse and null cases.

New adapters were stored separately. The original arms, base, and all recipe variants were regenerated together in each campaign. Tables retain every adverse/null outcome, paired intervals and training-seed ranges; no best-recipe selection supports the primary claim.

Completed measurements. The frozen prompt panel retains directional loss under forward SFT for translation and format conversion; unit conversion remains a null-collapse case under the original recipe (Table 20). Preservation is not uniform across tasks or recipes: the Python-C++ mixture has large seed-dependent forward costs, and the higher learning-rate unit recipe includes reverse loss as well as improvement (Table 21). The unit boundary case is therefore scoped to the original recipe. Every frozen prompt and recipe is retained; selecting the best variant does not support the pri-

Task	Model	Seed	Δ CL-fwd versus SFT	Δ CL-mix5 versus mix5
fnt	gemma3-4b	17	-2.88 [-3.88, -1.96]	+1.08 [+0.32, +1.88]
fnt	gemma3-4b	42	-1.28 [-1.96, -0.64]	-1.04 [-1.80, -0.32]
fnt	gemma3-4b	1234	+2.36 [+1.60, +3.20]	-0.32 [-0.92, +0.24]
fnt	llama31-8b	17	+2.24 [+1.52, +3.04]	+0.48 [-0.12, +1.12]
fnt	llama31-8b	42	-33.12 [-35.48, -30.64]	+0.40 [-0.20, +1.00]
fnt	llama31-8b	1234	+0.00 [+0.00, +0.00]	-0.72 [-1.52, +0.12]
fnt	llama32-3b	17	-0.24 [-0.56, +0.04]	+0.00 [-1.12, +1.12]
fnt	llama32-3b	42	-1.48 [-2.12, -0.92]	+1.68 [+0.52, +2.88]
fnt	llama32-3b	1234	-0.52 [-1.00, -0.04]	+4.44 [+3.08, +5.84]
mt_de-en	gemma3-12b	17	+0.00 [+0.00, +0.00]	+0.79 [-1.78, +3.56]
mt_de-en	gemma3-12b	42	-0.10 [-0.40, +0.00]	+0.59 [-2.08, +3.26]
mt_de-en	gemma3-12b	1234	+0.00 [+0.00, +0.00]	+0.69 [-1.68, +3.16]
mt_de-en	gemma3-4b	17	+0.00 [+0.00, +0.00]	+0.10 [-2.27, +2.27]
mt_de-en	gemma3-4b	42	+0.00 [+0.00, +0.00]	+0.40 [-2.08, +2.77]
mt_de-en	gemma3-4b	1234	+0.00 [+0.00, +0.00]	-0.30 [-2.47, +2.17]
mt_de-en	llama31-8b	17	-16.50 [-19.47, -13.64]	-1.19 [-4.05, +1.88]
mt_de-en	llama31-8b	42	-13.54 [-16.50, -11.07]	-0.40 [-3.26, +2.57]
mt_de-en	llama31-8b	1234	-23.32 [-26.78, -19.96]	-1.28 [-4.35, +1.58]
mt_de-en	llama32-3b	17	+0.00 [+0.00, +0.00]	+0.89 [-1.38, +3.26]
mt_de-en	llama32-3b	42	+0.00 [+0.00, +0.00]	+2.47 [+0.00, +5.04]
mt_de-en	llama32-3b	1234	-0.10 [-0.40, +0.00]	+0.10 [-2.57, +2.87]
units	gemma3-12b	17	+0.50 [-1.70, +2.80]	+0.70 [-0.90, +2.30]
units	gemma3-12b	42	-1.40 [-2.70, -0.20]	+0.40 [-1.10, +2.00]
units	gemma3-12b	1234	-4.90 [-7.00, -3.10]	+0.50 [-0.90, +1.90]

Table 10: Completed exploratory reverse-generation contrasts (percentage points), with 99% paired cluster-bootstrap intervals. Five primary contrasts are adjusted within each cell; no adjustment across cells. All completed original and scale/domain-extension comparisons are included; excluded cells remain eligibility boundaries rather than tuned comparisons.

mary claim. These seed ranges are descriptive, and endpoint scales are not pooled into an overall preservation score.

C.6 Initial task feasibility and full-weight checks

Proposed task comparisons require the base model to perform both directions and require correctness checks that accept reference answers while rejecting copies and invalid outputs. Table 22 states which requirements fail for the original small-model extensions. Excluding these comparisons avoids interpreting an initial floor or a scorer failure as fine-tuning damage. Original unit-conversion experiments at Gemma12B improve both directions under forward SFT; these repeated null results show that degradation is not inherent to every paired task. Python/C++ translation at Llama8B provides an additional successful starting-task comparison. The revised-contract results above reuse their underlying corpora; they are not new domains.

Full-weight fine-tuning compares selected Llama3B runs with forward-only and 5 training (Table 25). Each full checkpoint uses an isolated inference engine with identical held-out instances, prompts, tokenizer and scoring settings.

Small changes must be interpreted against the measured decoding floor across engine restarts. The separate saved-checkpoint spectral repair experiment concerns later repair, whereas replacement mixtures preserve during training. Its improvement in both directions is a scoped result and does not establish a general internal cause.

C.7 Result selection and reproducibility

All completed comparisons and null/adverse outcomes are retained. Repeated evaluations use the latest complete evaluation, independently of effect sign. Revised scorers and prompts are separate exploratory measurements, with their original failures preserved; differences across versions are not paired treatment effects. Test-pair bootstrap uncertainty is reported separately from variation across training seeds. An interval spanning zero does not show equivalence, and a degenerate interval at a generation floor needs particular caution.

The accompanying data record fixed corpora, prompts, scorer sources, training configurations, per-instance generations and analysis commands. Hashes and complete paired coverage are checked before table generation; failed adapter loading excludes an evaluation. Some historical training code and upstream dataset revisions were not re-

Task	Model	Seed	CL-fwd vs extra CE	CL-fwd vs shuffled	CL-mix5 vs extra CE
fnt	Gemma 4B	17	+0.04 [-0.84, +0.96]	-29.20 [-31.80, -26.72]	-0.08 [-0.72, +0.56]
fnt	Gemma 4B	42	-16.04 [-18.00, -14.20]	-19.44 [-21.52, -17.48]	-0.44 [-1.28, +0.44]
fnt	Gemma 4B	1234	+8.20 [+6.80, +9.64]	-22.16 [-25.08, -19.24]	-0.60 [-1.24, +0.00]
fnt	llama31-8b	17	+2.24 [+1.52, +3.04]	+2.24 [+1.52, +3.04]	+0.44 [-0.16, +1.00]
fnt	llama31-8b	42	-21.52 [-23.64, -19.44]	-33.00 [-35.40, -30.52]	+0.16 [-0.32, +0.68]
fnt	llama31-8b	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.48 [-1.24, +0.28]
fnt	Llama 3B	17	-0.48 [-0.96, -0.08]	-0.68 [-1.16, -0.24]	+1.76 [+0.56, +2.96]
fnt	Llama 3B	42	-1.76 [-2.48, -1.12]	-3.40 [-4.32, -2.52]	+0.64 [-0.44, +1.68]
fnt	Llama 3B	1234	-0.36 [-0.88, +0.12]	+0.28 [+0.08, +0.60]	+2.76 [+1.48, +4.00]
mt_de-en	gemma3-12b	17	+0.10 [+0.00, +0.40]	-0.10 [-0.40, +0.00]	+0.49 [-2.17, +2.96]
mt_de-en	gemma3-12b	42	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	-1.19 [-3.56, +1.19]
mt_de-en	gemma3-12b	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.40 [-2.87, +2.08]
mt_de-en	Gemma 4B	17	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.59 [-2.77, +1.58]
mt_de-en	Gemma 4B	42	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [-2.27, +2.27]
mt_de-en	Gemma 4B	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+1.09 [-1.09, +3.36]
mt_de-en	llama31-8b	17	-18.18 [-21.34, -15.12]	+0.00 [+0.00, +0.00]	-1.88 [-5.04, +1.28]
mt_de-en	llama31-8b	42	-14.53 [-17.49, -11.96]	-0.20 [-0.59, +0.00]	-0.40 [-3.26, +2.47]
mt_de-en	llama31-8b	1234	-21.74 [-25.10, -18.38]	-0.10 [-0.49, +0.00]	-1.09 [-4.15, +1.68]
mt_de-en	Llama 3B	17	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+0.69 [-1.68, +3.16]
mt_de-en	Llama 3B	42	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+1.78 [-0.59, +4.25]
mt_de-en	Llama 3B	1234	-0.10 [-0.40, +0.00]	-0.10 [-0.40, +0.00]	-0.69 [-3.26, +1.88]
units	gemma3-12b	17	+11.60 [+8.50, +14.90]	-3.80 [-5.60, -2.00]	+0.50 [-1.00, +2.00]
units	gemma3-12b	42	-1.10 [-2.50, +0.20]	-0.10 [-1.80, +1.60]	+1.00 [-0.50, +2.60]
units	gemma3-12b	1234	-2.60 [-4.60, -0.80]	-4.80 [-7.00, -2.70]	+0.80 [-0.60, +2.30]

Table 11: The remaining three registered primary reverse-generation contrasts, in percentage points with 99% paired cluster-bootstrap intervals. Together with Table 10, these display all five primary comparisons for every completed cell. Extra CE is an exposure/compute proxy, not matched measured FLOPs. Degenerate zero intervals at an observed generation floor do not establish equivalence. These intervals are exploratory.

tained, so reproducible result analysis does not imply exact historical training reconstruction. These missing versions are explicit limitations, rather than inferred from a current cached dataset.

D Likelihood, candidate selection and layer interventions

Five experiments study candidate likelihood, local gradients and layer-specific interference. The subset was chosen before these diagnostic outcomes to cover distinct questions within the available computational budget. These single-seed analyses are exploratory; paired intervals measure test-pair uncertainty, not training-seed stability. All candidates, controls, bands and local steps remain in the evidence bundle. Completed diagnostic experiments: 5. In the completed Llama formatting and translation diagnostics, forward SFT lowers gold completion loss in the forward direction and raises it in reverse; mixed-direction SFT lowers reverse loss relative to SFT with small mean forward changes (Table 26). The local SFT gradient cosines are near zero (Table 27); these checkpoints provide little support for a simple persistent gradient-conflict account. Gradients at these final checkpoints do not reconstruct interference

along the training trajectory. This descriptive evidence concerns answer likelihood and does not establish stored knowledge. The Gemma translation diagnostic differs: reverse gold completion loss improves relative to base despite collapsed free generation. Absolute answer loss therefore does not uniformly track collapse. In the three collapse cells, per-token gold-over-copy preference falls under forward SFT and recovers under mixed-direction SFT; the units null cell retains strong preference (Table 28). Total and per-token rankings are reported together because candidate lengths differ. Translation has multiple valid answers; a restricted candidate ranking is neither correctness nor a measure of retained knowledge. At the SFT checkpoint, both tested reverse and mixed local-step sizes lower reverse held-out loss in every cell, including the null cell; this is not a collapse-specific signature or a demonstrated generation repair. All checkpoint states, update directions, step sizes and both endpoints are retained in the bundle. Formatting middle-band removals improve reverse success beyond all matched controls with small forward changes, whereas the earliest band harms forward performance without consistent reverse recovery (Table 29). This supports

Task/model	Objective	Compared with	Forward change	Backward change
English → German/L8B	Unlikelihood	Forward SFT	+0.9 [-1.2, +2.9]	-58.6 [-61.5, -55.5]
English → German/L8B	Unlikelihood	5% reversal	+1.0 [-1.2, +3.2]	-69.1 [-71.8, -66.3]
English → German/L3B	Unlikelihood	Forward SFT	+0.3 [-1.6, +2.1]	-63.6 [-66.4, -60.6]
English → German/L3B	Unlikelihood	5% reversal	-0.3 [-2.3, +1.7]	-68.8 [-71.6, -65.9]
Code transformation/L3B	CFT	Forward SFT	+2.8 [+1.1, +4.4]	+3.1 [+1.8, +4.2]
Code transformation/L3B	CFT	5% reversal	-0.1 [-1.7, +1.5]	-68.7 [-71.1, -66.3]
Question → SQL/L8B	Round-trip	Forward SFT	-0.8 [-2.2, +0.6]	+7.5 [+5.0, +10.0]
Question → SQL/L8B	Round-trip	5% reversal	-0.6 [-2.3, +1.2]	+1.0 [-1.1, +3.0]
Question → SQL/L3B	Round-trip	Forward SFT	-0.8 [-2.2, +0.8]	+24.9 [+21.7, +28.0]
Question → SQL/L3B	Round-trip	5% reversal	+0.5 [-1.2, +2.1]	-7.1 [-9.5, -4.8]

Table 12: Auxiliary-objective differences from forward-only and reversed-pair SFT, in percentage points with 95% paired cluster-bootstrap intervals. Every completed compatible objective comparison is included. Intervals are exploratory and unadjusted for multiplicity; small point-estimate gains do not establish an objective-specific benefit.

Evaluation	Model/task combinations	Training seeds	Runs
General probes	3	17,42,1234	9
Prompt/template tests	4	17,42,1234	12
Independent transfer	2	17,42,1234	6
Recipe evaluations	2	17,42,1234	6

Table 13: Additional evaluation design. Probe corpora contain 541 IFEval prompts and 1319 GSM8K questions; independent transfer retains 1688 OPUS translation pairs. Synthetic magnitude/template tests retain 1000 instances each. Recipe experiments compare 48 separately trained variants. Numbers describe corpus sizes and the fixed experimental design; results appear in the corresponding tables.

depth selectivity in one cell, with unadjusted intervals across multiple interventions, rather than a universal causal mechanism.

Task orientation	Train	Val.	Test	Retained	Eval.	Steps	Batch $\times$ accum.
German $\rightarrow$ English	6500	500	1012	6500	1012	306	16×4
English $\rightarrow$ German	6500	500	1012	6500	1012	306	4×16
English $\rightarrow$ Chinese	6500	500	1012	6500	1012	306	4×16
Chinese $\rightarrow$ English	6500	500	1012	6500	1012	306	4×16
Code transformation	6500	500	2060	6499	2060	306	4×16
Question $\rightarrow$ SQL	6499	500	1034	6499	1034	306	16×4
Format conversion	6500	500	2500	6500	2500	306	16×4
Factual relations	504	69	135	504	135	24	16×4
Polynomial expansion	6500	500	2500	6500	2500	306	4×16
Polynomial factorization	6500	500	2500	6500	2500	306	4×16
Execution prediction	302	100	200	302	200	15	16×4

Table 14: Built paired-corpus sizes and recorded forward-SFT training/evaluation counts for the main-table reference campaigns. Train/validation/test counts are verified against the current frozen JSONL files; retained rows, steps and batch shapes come from each saved training manifest/summary. These are distinct provenance records: the code campaign retains fewer training rows than the current built corpus. Historical training data bytes are not inferred from a current file hash. Eval. counts paired instances per arm and direction. Opposite task orientations reuse task families; they are not independent datasets.

Task	Model	Seed	Arm	Forward (%)	Reverse (%)
mt_de-en	llama32-3b	17	base	73.7	79.2
mt_de-en	llama32-3b	17	sft	67.1	0.0
mt_de-en	llama32-3b	17	replay	68.6	59.0
mt_de-en	llama32-3b	17	mix5	67.4	82.0
mt_de-en	llama32-3b	17	mix50	67.4	82.9
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	mix50	94.9	90.9

Table 15: Representative seed-17 preservation rates. Each model/task row set comes from one complete campaign. Additional doses, seed variation, paired intervals and measured cost records are retained in the source-backed analysis. The units boundary case is included independently of effect sign.

Task	Model	Arm	Minutes	Steps	CE corpus tokens
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	mix50	9.0	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	cl_fwd	241.9	306	1096262
mt_de-en	llama32-3b	cl_mix5	62.2	306	1096262
mt_de-en	llama32-3b	sft_extra_ce	123.2	306	1096262
mt_de-en	llama32-3b	mix5_extra_ce	119.1	306	1096262
fnt	gemma3-4b	sft	30.6	306	1927990
fnt	gemma3-4b	mix5	30.7	306	1927882
fnt	gemma3-4b	replay	30.2	306	1927978
fnt	gemma3-4b	cl_fwd	194.3	306	1927990
fnt	gemma3-4b	cl_mix5	193.5	306	1927882
fnt	gemma3-4b	sft_extra_ce	191.3	306	1927990
fnt	gemma3-4b	mix5_extra_ce	194.0	306	1927882
units	gemma3-12b	sft	21.6	306	788819
units	gemma3-12b	mix50	22.1	306	788819
units	gemma3-12b	replay	21.5	306	799547
units	gemma3-12b	sft	21.6	306	788819
units	gemma3-12b	mix5	15.6	306	788819
units	gemma3-12b	replay	21.5	306	799547
units	gemma3-12b	cl_fwd	82.8	306	788819
units	gemma3-12b	cl_mix5	83.0	306	788819
units	gemma3-12b	sft_extra_ce	166.9	306	788819
units	gemma3-12b	mix5_extra_ce	168.0	306	788819

Table 16: Recorded seed-17 trainer runtimes (minutes), optimizer steps, and sequence tokens in the tokenized CE corpus (before epoch repetition). Auxiliary encoded-side exposure remains separately recorded in the analysis; these token counts are not total objective FLOPs. Runtime varies with GPU/training layout and is not a claim of matched wall time. Missing measurements are unavailable, never inferred.

Cell	Endpoint	Seeds	SFT-base	mix5-SFT	Replay-SFT
fnt/Gemma4B	gsm8k	3	[-56.6, -36.3]	[+0.5, +21.8]	[+24.7, +46.0]
fnt/Gemma4B	ifeval	3	[-1.3, +0.6]	[-3.0, +0.9]	[-8.3, -4.1]
mt_de-en/Llama3B	gsm8k	3	[+6.6, +8.7]	[-2.7, +1.4]	[-13.7, +10.6]
mt_de-en/Llama3B	ifeval	3	[-10.9, -9.6]	[-0.4, +4.4]	[+0.2, +3.0]
units/Gemma12B	gsm8k	3	[+1.2, +2.0]	[-0.7, +0.9]	[-10.4, -9.1]
units/Gemma12B	ifeval	3	[-3.0, +1.5]	[+0.0, +3.5]	[-13.5, -3.0]

Table 17: Strict general-ability endpoint changes: descriptive minimum/maximum across all completed training seeds, in percentage points. Each difference uses its own fresh campaign baseline. These ranges are not confidence intervals. Per-seed paired intervals, all objective controls and overlap exclusions remain in the evidence artifact; endpoints are not pooled.

Model	Seed	Direction	Base (%)	Format failure (%)	Echo (%)	Campaign
gemma3-4b	1234	forward	33.9	10.1	6.6	blocked
gemma3-4b	1234	reverse	44.9	6.3	0.0	blocked
gemma3-4b	17	forward	33.9	10.0	6.6	blocked
gemma3-4b	17	reverse	44.8	6.3	0.0	blocked
gemma3-4b	42	forward	33.9	10.1	6.6	blocked
gemma3-4b	42	reverse	44.8	6.3	0.0	blocked
llama32-3b	1234	forward	33.7	10.6	6.8	blocked
llama32-3b	1234	reverse	53.0	6.8	0.0	blocked
llama32-3b	42	forward	33.7	10.6	6.8	blocked
llama32-3b	42	reverse	52.9	6.8	0.0	blocked
llama32-3b	17	forward	33.7	10.6	6.8	blocked
llama32-3b	17	reverse	52.8	6.8	0.0	blocked

Table 18: Completed OPUS transfer eligibility measurements with unchanged thresholds. Each campaign requires both directions to satisfy the frozen rate, format-failure, and echo conjunction. Failed campaigns preserve base-only trials and do not generate tuned comparisons; an echo failure is a criterion-validity boundary, not evidence of adapter collapse.

Versioned task	Model	Seed	Dir.	Base	SFT	50%	Replay	Rev.
Formatting contract	L8B	17	F	34.4	99.9	99.9	99.9	0.0
			B	60.0	63.0	100.0	54.8	99.9
Formatting contract	L8B	42	F	34.9	99.9	99.9	99.7	0.1
			B	60.9	61.6	100.0	16.5	99.9
Formatting contract	L8B	1234	F	34.4	99.9	100.0	99.7	1.1
			B	60.0	66.6	100.0	15.7	99.9
Units contract	G12B	17	F	39.9	97.0	95.1	92.2	38.7
			B	46.9	70.3	89.9	65.0	95.7
Units contract	G12B	42	F	39.9	97.3	93.8	92.9	27.2
			B	46.9	73.3	88.0	73.0	95.6
Units contract	G12B	1234	F	39.9	97.2	94.4	94.4	9.1
			B	46.9	71.2	88.6	70.4	95.9
Typed Python/C++	G12B	17	F	86.2	100.0	100.0	9.2	0.0
			B	95.0	35.0	100.0	0.0	100.0
Typed Python/C++	G12B	42	F	86.2	100.0	85.0	36.1	0.0
			B	95.0	72.7	100.0	30.5	100.0
Typed Python/C++	G12B	1234	F	86.2	74.8	73.5	2.3	0.0
			B	95.0	55.8	100.0	0.0	100.0
Typed Python/C++	L8B	17	F	84.2	100.0	100.0	99.6	0.0
			B	100.0	0.0	48.4	99.4	0.0
Typed Python/C++	L8B	42	F	85.9	100.0	1.8	99.9	0.0
			B	100.0	0.0	0.0	99.4	0.0
Typed Python/C++	L8B	1234	F	85.9	100.0	100.0	95.9	0.0
			B	100.0	0.0	0.1	97.8	0.0

Table 19: Revised-output task comparisons, strict success (%). Each F/B pair uses its own base evaluated on the same examples and all controls; training seeds are reported separately. These post-inspection contracts reuse the original semantic corpora and are exploratory diagnostics, not independent task replication. Only the 50% reversal dose was tested here. Red uses the descriptive SFT backward-loss threshold from the main table. Original failures remain reported separately; no cross-version subtraction or best-seed selection is used.

Cell	Setting	Seeds	ΔF_{S-B}	ΔR_{S-B}	ΔF_{M-S}	ΔR_{M-S}
fmt/G4B	original/contract_first	3	[+20.0, +20.2]	[-58.5, -27.8]	[+0.0, +0.1]	[+51.9, +82.1]
fmt/G4B	original/primary	3	[+21.4, +21.7]	[-60.0, -27.9]	[+0.0, +0.2]	[+51.8, +83.8]
fmt/G4B	original/reword	3	[+22.2, +22.2]	[-59.8, -27.8]	[-0.0, +0.1]	[+51.8, +83.5]
mt_de-en/L3B	original/contract_first	3	[-4.4, -4.2]	[-79.0, -78.7]	[-0.5, +0.6]	[+79.9, +81.1]
mt_de-en/L3B	original/primary	3	[-7.7, -5.5]	[-79.7, -79.6]	[-0.3, +0.7]	[+79.2, +82.0]
mt_de-en/L3B	original/reword	3	[-4.6, -3.1]	[-79.1, -78.9]	[-0.7, +0.5]	[+78.8, +81.6]
py_cpp/L8B	original/contract_first	3	[+4.4, +4.4]	[-100.0, -100.0]	[-92.5, +0.0]	[+0.0, +50.0]
py_cpp/L8B	original/primary	3	[+15.8, +15.8]	[-100.0, -100.0]	[-97.8, +0.0]	[+0.0, +48.7]
py_cpp/L8B	original/reword	3	[+6.4, +6.4]	[-100.0, -100.0]	[-89.2, +0.0]	[+0.0, +49.9]
py_cpp/L8B	py_cpp_pieewise_quadratic/primary	3	[-68.2, -68.2]	[+0.0, +0.0]	[+0.0, +0.0]	[+99.3, +100.0]
units/G12B	original/contract_first	3	[+63.6, +63.6]	[+32.5, +36.6]	[-2.3, -1.4]	[+17.1, +20.7]
units/G12B	original/primary	3	[+63.3, +63.3]	[+34.0, +38.0]	[-2.1, -1.4]	[+17.4, +21.3]
units/G12B	original/reword	3	[+65.3, +65.6]	[+47.3, +50.7]	[-2.3, -1.6]	[+17.4, +20.6]
units/G12B	units_magnitude/primary	3	[+35.1, +47.2]	[+36.7, +40.0]	[-1.4, +5.0]	[+6.0, +8.5]

Table 20: Completed robust panel: ranges across training seeds of within-campaign strict-success changes (percentage points). S , B , and M denote forward SFT, the base evaluated on the same examples, and the directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

Cell	Setting	Seeds	ΔF_{S-B}	ΔR_{S-B}	ΔF_{M-S}	ΔR_{M-S}
fmt/L3B	lr_high	3	[+39.5, +39.5]	[-51.1, -49.2]	[-0.2, -0.0]	[+95.4, +98.2]
fmt/L3B	lr_low	3	[+38.7, +39.1]	[-43.0, -40.6]	[+0.2, +0.6]	[+82.6, +86.6]
fmt/L3B	original	3	[+39.4, +39.5]	[-50.8, -49.8]	[-0.1, +0.2]	[+92.2, +95.7]
fmt/L3B	rank_high	3	[+39.0, +39.3]	[-51.1, -51.0]	[-0.0, +0.4]	[+95.8, +96.6]
fmt/L3B	rank_low	3	[+38.2, +39.4]	[-50.4, -45.4]	[-0.0, +1.1]	[+90.8, +92.9]
units/G12B	lr_high	3	[+63.6, +63.7]	[-28.2, +12.1]	[-4.2, -1.1]	[+43.7, +84.6]
units/G12B	lr_low	3	[+62.8, +63.1]	[+36.1, +37.4]	[-3.4, -2.5]	[+14.1, +16.2]
units/G12B	original	3	[+63.3, +63.6]	[+34.1, +37.8]	[-2.3, -1.2]	[+17.7, +21.2]
units/G12B	rank_high	3	[+63.4, +64.3]	[+13.7, +27.9]	[-2.7, -1.9]	[+27.7, +41.1]
units/G12B	rank_low	3	[+62.8, +63.6]	[+25.6, +37.6]	[-3.1, -0.4]	[+15.9, +28.3]

Table 21: Completed recipe panel: ranges across training seeds of within-campaign strict-success changes (percentage points). S , B , and M denote forward SFT, the base evaluated on the same examples, and the directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

Task	Model	Reason an original comparison is excluded
units	L3B	forward success below 10%; forward format failures above 15%; backward success below 10%; backward format failures above 15%
units	G4B	forward format failures above 15%; backward success below 10%; backward format failures above 15%
logic	L3B	forward success below 10%; forward format failures above 15%; backward success below 10%; backward format failures above 15%
logic	G4B	forward success below 10%; forward format failures above 15%; backward success below 10%
py_cpp	L3B	forward format failures above 15%; backward format failures above 15%
py_cpp	G4B	forward success below 10%; forward format failures above 15%

Table 22: Scientific reasons for excluding the original small-model task extensions. Correctness criteria are unchanged; revised-output versions are separate exploratory comparisons.

Task	Model	Forward	Reverse	Eligibility
units	llama32-3b	0.4	1.9	Fail
units	gemma3-4b	37.6	3.5	Fail
logic	llama32-3b	0.0	0.0	Fail
logic	gemma3-4b	0.0	0.0	Fail
py_cpp	llama32-3b	78.6	74.1	Fail
py_cpp	gemma3-4b	5.1	99.9	Fail

Table 23: Initial task feasibility: base-model strict success in percent. Comparisons require adequate performance in both directions and a scorer that rejects copied outputs. Explicit-output diagnostics use the same semantic splits and are separate exploratory variants.

Task	Model	Seed	Arm	Forward	Reverse
py_cpp	llama31-8b	17	base	85.9	100.0
py_cpp	llama31-8b	17	sft	100.0	0.0
py_cpp	llama31-8b	17	mix50	100.0	48.7
py_cpp	llama31-8b	17	replay	100.0	97.6
py_cpp	llama31-8b	17	rev	0.0	0.0
py_cpp	llama31-8b	42	base	84.2	100.0
py_cpp	llama31-8b	42	sft	100.0	0.0
py_cpp	llama31-8b	42	mix50	1.7	0.0
py_cpp	llama31-8b	42	replay	99.9	99.5
py_cpp	llama31-8b	42	rev	0.0	0.0
units	gemma3-12b	1234	base	33.7	35.8
units	gemma3-12b	1234	sft	96.9	72.7
units	gemma3-12b	1234	mix50	95.1	90.7
units	gemma3-12b	1234	replay	94.8	71.9
units	gemma3-12b	1234	rev	47.3	94.7
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	mix50	94.9	90.9
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	rev	44.0	95.0
units	gemma3-12b	42	base	33.7	35.8
units	gemma3-12b	42	sft	97.3	73.7
units	gemma3-12b	42	mix50	95.7	91.3
units	gemma3-12b	42	replay	95.5	72.7
units	gemma3-12b	42	rev	45.9	94.6
py_cpp	llama31-8b	1234	base	84.2	100.0
py_cpp	llama31-8b	1234	sft	100.0	0.0
py_cpp	llama31-8b	1234	mix50	100.0	0.2
py_cpp	llama31-8b	1234	replay	95.8	97.9
py_cpp	llama31-8b	1234	rev	0.0	0.0

Table 24: Larger-model task comparisons: strict success in percent, with all adaptations and the base evaluated on the same examples per training run. Unit-conversion replications improve both directions under SFT, providing a null case for directional collapse.

Task	Seed	Direction	Base	Full SFT	Full mix5
code	17	forward	37.9	70.0	70.7
code	17	reverse	31.9	2.0	65.9
mt_de-en	17	forward	73.3	70.9	72.0
mt_de-en	17	reverse	80.1	72.8	83.6
mt_en-de	17	forward	80.1	87.0	86.6
mt_en-de	17	reverse	72.9	69.9	72.4
mt_en-zh	17	forward	73.9	79.1	79.2
mt_en-zh	17	reverse	75.9	66.9	71.8
mt_zh-en	17	forward	76.3	74.2	73.9
mt_zh-en	17	reverse	73.8	70.2	78.5
sql	17	forward	49.3	60.5	59.7
sql	17	reverse	50.4	18.7	55.1

Table 25: Llama3B full-weight fine-tuning, with forward and backward success in percent. Checkpoints use isolated engines and identical held-out instances; these comparisons span engine restarts. Paired intervals accompany the result data.

Task	Model	Direction	SFT minus base	mix5 minus SFT
fmt	llama32-3b	forward	-0.117 [-0.132, -0.101]	+0.000 [-0.000, +0.001]
fmt	llama32-3b	reverse	+0.121 [+0.104, +0.139]	-0.288 [-0.331, -0.247]
mt_de-en	llama32-3b	forward	-0.277 [-0.336, -0.220]	+0.004 [-0.003, +0.012]
mt_de-en	llama32-3b	reverse	+0.103 [+0.075, +0.131]	-0.181 [-0.206, -0.155]
mt_de-en	gemma3-4b	forward	-2.352 [-2.547, -2.168]	-0.002 [-0.012, +0.008]
mt_de-en	gemma3-4b	reverse	-1.008 [-1.122, -0.893]	-0.419 [-0.459, -0.378]
units	gemma3-12b	forward	-2.587 [-2.829, -2.358]	+0.016 [-0.005, +0.053]
units	gemma3-12b	reverse	-2.612 [-2.832, -2.405]	-0.381 [-0.503, -0.273]

Table 26: Gold completion NLL per token, paired differences with descriptive 95% bootstrap intervals on 128 fixed pairs. Negative values favor the first system. Candidate likelihood does not establish generation correctness or stored knowledge. All seeds here are 17.

Task	Model	SFT	Replay	mix5
fmt	llama32-3b	-0.018	+0.004	+0.010
mt_de-en	llama32-3b	-0.000	+0.048	+0.059
mt_de-en	gemma3-4b	+0.021	+0.012	+0.031
units	gemma3-12b	-0.013	+0.009	-0.026

Table 27: Forward/reverse validation-gradient cosine on 16 pairs in local LoRA A/B coordinates. Gemma excludes inactive vision factors. These coordinate-dependent quantities do not by themselves establish interference or causality.

Task	Model	Direction	Base	SFT	mix5
fmt	llama32-3b	forward	0.695 / 0.719	1.000 / 1.000	1.000 / 1.000
fmt	llama32-3b	reverse	0.750 / 0.672	0.000 / 0.008	1.000 / 1.000
mt_de-en	llama32-3b	forward	0.344 / 0.195	0.484 / 0.234	0.445 / 0.242
mt_de-en	llama32-3b	reverse	0.062 / 0.180	0.000 / 0.000	0.320 / 0.562
mt_de-en	gemma3-4b	forward	0.375 / 0.289	0.461 / 0.398	0.359 / 0.273
mt_de-en	gemma3-4b	reverse	0.406 / 0.586	0.000 / 0.000	0.430 / 0.562
units	gemma3-12b	forward	0.828 / 0.875	1.000 / 1.000	0.992 / 1.000
units	gemma3-12b	reverse	0.586 / 0.547	1.000 / 0.977	1.000 / 1.000

Table 28: Fraction of 128 fixed pairs where gold completion NLL is strictly lower than copied-input NLL: total / per-token ranking. Candidate lengths differ, so these rankings need not agree. Both directions and the same focal systems as the likelihood contrasts are shown; all eight systems and shuffled-target candidates remain in the bundle. Single training seed 17.

Removed band	Matched control	Direction	Success difference
band0_remove	band0_scale	forward	-0.043 [-0.062, -0.025]
band0_remove	band0_scale	reverse	-0.006 [-0.014, +0.000]
band0_remove	band0_random101	forward	-0.039 [-0.057, -0.021]
band0_remove	band0_random101	reverse	-0.014 [-0.023, -0.004]
band0_remove	band0_random202	forward	-0.037 [-0.055, -0.020]
band0_remove	band0_random202	reverse	+0.000 [+0.000, +0.000]
band1_remove	band1_scale	forward	-0.010 [-0.021, +0.000]
band1_remove	band1_scale	reverse	+0.096 [+0.070, +0.121]
band1_remove	band1_random101	forward	-0.008 [-0.018, +0.000]
band1_remove	band1_random101	reverse	+0.094 [+0.068, +0.119]
band1_remove	band1_random202	forward	-0.008 [-0.018, +0.000]
band1_remove	band1_random202	reverse	+0.100 [+0.074, +0.125]
band2_remove	band2_scale	forward	-0.002 [-0.012, +0.006]
band2_remove	band2_scale	reverse	+0.084 [+0.061, +0.107]
band2_remove	band2_random101	forward	+0.000 [-0.010, +0.010]
band2_remove	band2_random101	reverse	+0.082 [+0.057, +0.107]
band2_remove	band2_random202	forward	+0.002 [-0.008, +0.012]
band2_remove	band2_random202	reverse	+0.088 [+0.062, +0.113]
band3_remove	band3_scale	forward	+0.002 [+0.000, +0.006]
band3_remove	band3_scale	reverse	+0.037 [+0.021, +0.055]
band3_remove	band3_random101	forward	+0.006 [+0.000, +0.014]
band3_remove	band3_random101	reverse	+0.033 [+0.018, +0.051]
band3_remove	band3_random202	forward	+0.008 [+0.002, +0.016]
band3_remove	band3_random202	reverse	+0.043 [+0.025, +0.061]

Table 29: All fixed-depth removals versus retained-norm-matched controls for formatting/Llama3B (seed17), strict no-echo success differences on 512 pairs with descriptive 95% paired intervals. Retained and edit norms are both recorded; only retained norms are matched. No best band or control is selected.